Miroslav Šimko, PhD

Machine Learning / Python / C++ Developer
Currently based in Kyoto, Japan

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Professional Summary

Machine-learning developer with a Ph.D. in experimental nuclear and particle physics. Developing road-marking and pavement segmentation with AI coding agents (work in progress), alongside 3D point-cloud processing at ioLabs. Earlier work includes a deployed Mask R-CNN plan-stamping system and heavy-flavor analysis, detector calibration and operations at STAR.

Education

2013/10–2021/03, Ph.D. (defence)
Experimental Nuclear and Particle Physics
Czech Technical University in Prague (CTU)
Faculty of Nuclear Sciences and Physical Engineering (FNSPE)
Thesis focus: Study of Heavy Flavor at the STAR Experiment

Autumn 2011–2013/09, Ing./M.Sc.
Experimental Nuclear and Particle Physics (CTU – FNSPE)
Thesis: Design and Optimization of the Optical Readout System for Electromagnetic Calorimeter FOCAL for the ALICE Experiment

Autumn 2008–2011/09, B.Sc.
Experimental Nuclear and Particle Physics (CTU – FNSPE)
Thesis: Detector Control System for the ALICE Experiment

Experience

2021–present: ioLabs
Developer (Machine Learning, Python, C++)
- Developing road-marking (line) and pavement (asphalt) segmentation with U-Net and PyTorch, using AI coding agents (work in progress). Training and inference packages use Azure ML and DVC.
- Contribute to Kirioll, a team project converting motorway laser scans into CAD data: 3D point-cloud processing and photo/point-cloud fusion.
- Built a Mask R-CNN plan-stamping system with a classical-algorithm fallback, still deployed through FastAPI; also used AWS Textract.
- Developed, trained and deployed computer-vision models on polygon meshes; deployed applications with Flask, FastAPI, Docker and Jenkins.
- Used GPT-3.5 and GPT-4 for technical-document work; supported CAD-plugin work with small C++ changes.
- Quantized and retrained a Mask R-CNN experimentally; the quantized model was not used in the end.

2019/02–2019/12: ioLabs s.r.o. (agreement term)
Programming and scripting (C++, Python, C#)
- Limited-hours agreement alongside Ph.D. research.

Research Experience

2013–2021: Ph.D. research in particle physics / data science
Brookhaven National Laboratory, USA: STAR research, 2013–2021.
Lawrence Berkeley National Laboratory, USA: affiliate appointments within 2013/11–2019/05.
Nuclear Physics Institute, Czech Academy of Sciences: 2015–2021.

- Led a Λ_c analysis at STAR in close collaboration with colleagues: analysis code, topological-cut optimization and mixed-event background software (2014–2021), using ROOT and C++.
- STAR ZDC operations, calibration and on-call responsibility (2015–2020).
- Evaluated the PXL slow simulator (2013–2014); worked on signal processing, statistics and tabular prediction.
- Advised a bachelor's thesis and master's students.

2011–2013: CERN, Switzerland
Detector-Control Systems Developer (PVSS, C/C++)

- Developed and maintained detector remote-control systems and contributed to a new particle-detector prototype.

Languages and Skills

Czech/Slovak (native), English (fluent), French (beginner). Japanese: intermediate, self-assessed around JLPT N3, uncertified.

Python, C/C++, Bash, PyTorch, U-Net, Mask R-CNN, Open3D, ROOT, statistical analysis.

Linux, Docker, Jenkins, Git, Flask, FastAPI; Azure ML, DVC and AWS Textract through own project work.

Limited C# and FPGA experience (small changes in existing codebases). GPU work through PyTorch; no direct CUDA programming.

Interests

Physics, Automation, Photography, Sports, Reading

Publications and Conferences

Published article in Physical Review Letters, author on more than 95 collaboration papers.
Nine conference and workshop presentations, including talks and posters.

References

Martin Loučka (current employer)
ioLabs AG
CEO
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Zhangbu Xu (Work on STAR ZDC)
BNL, USA
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Jaroslav Bielčík (Former supervisor)
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Xin Dong (Supervisor at LBNL)
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